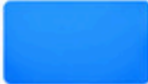


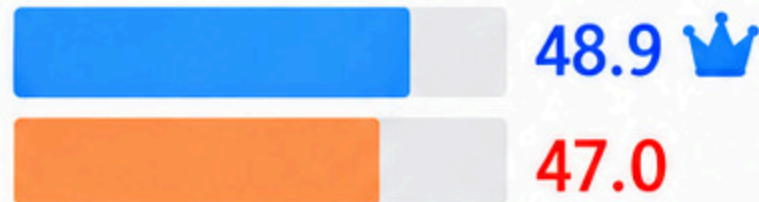
Criterion satisfaction (SOR)

GRPO: better criterion satisfaction

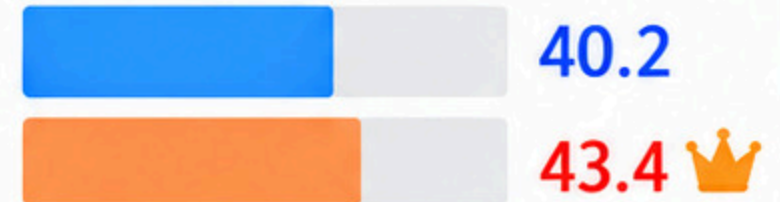
 GRPO  GDPO

IND

Mistral
IND

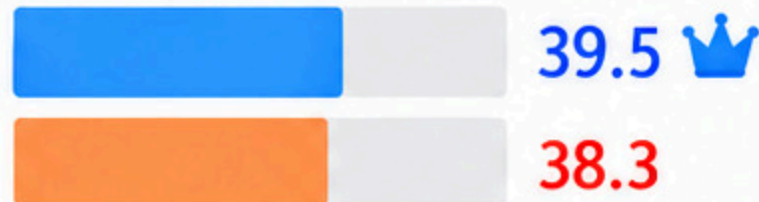


LLaMA
IND

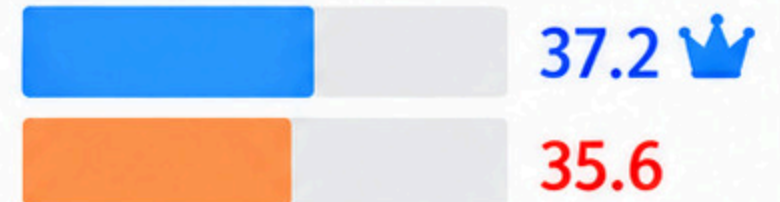


OOD

Mistral
OOD



LLaMA
OOD



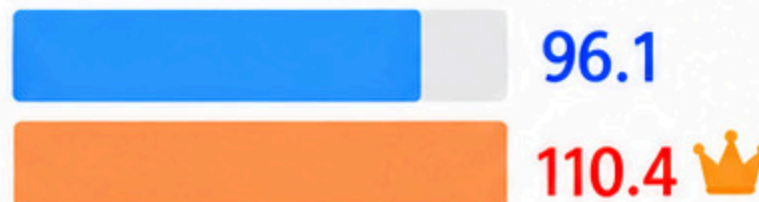
Relative improvement (RI)

GDPO: stronger relative improvement

 GRPO  GDPO

IND

Mistral
IND



LLaMA
IND



OOD

Mistral
OOD



LLaMA
OOD

